

PAPER ECH-Sustained Thermal-Barrier Decoupling in the D–³He Tandem-Mirror Plug: Hot-Electron and Sloshing-Ion Formation of the Ion-Confining Potential¹

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Abstract

The recirculating power of a tandem-mirror fusion burner is set almost entirely by the electrostatic potential the end plugs must raise to confine the central-cell ions, and in a deuterium–helium-3 (D–³He) machine that potential is large: the frozen Kronos burner design point demands an ion-confining potential $\phi_i \simeq 248.75$ keV at an ion temperature $T_i \simeq 90$ keV ($e\phi_i/T_i = 2.764$), a mirror ratio $R_{mc} = 4.61$, and an engineering gain $Q_E = 1.318$ that sits only just above break-even. The binding constraint is that, when the plug electrons and the central-cell electrons are Boltzmann-connected along the field line, raising ϕ_i requires raising the central-cell electron temperature, and heating the whole central column to hold the potential costs a frozen 1092.5 MW of electron-cyclotron power. We formalise the *thermal-barrier* mechanism that breaks this coupling. A localised depression of the electron density and potential, sustained jointly by electron-cyclotron-heated (ECH) hot electrons and a sloshing-ion population whose turning points bracket the barrier, thermally isolates the plug electrons from the central-cell electrons, so the confining potential is set by a hot *plug* electron temperature and a density ratio rather than by the central-cell temperature. We derive the field-line potential structure, the barrier-depth decoupling condition $e\phi_b/T_e \geq \ln(n_c/n_b)$, the sloshing-ion density profile from the adiabatic invariants, and the Pastukhov confinement scaling $n\tau_p \propto (e\phi_i/T_i) e^{e\phi_i/T_i}$, and we solve the coupled quasineutrality–confinement–power balance self-consistently. Booking the barrier, the confining-potential requirement falls to $\phi_i \simeq 167.9$ keV — a 33% cut in the binding gate, dominated by making T_i the lever — and the 1092.5 MW of central-cell heating is removed from the recirculating budget, lifting Q_E from 1.318 toward 3–5 in the fully realised limit. The kinetic barrier model (bounce-averaged Fokker–Planck) and the loss-cone microstability bound (warm-fill knee at 6.25%; residual Alfvén ion-cyclotron activity at $\gamma/\omega_{ci} \simeq 0.037$) are computed on the NVIDIA GPU high-performance-computing campaign. Against the TMX-U, GAMMA-10 and gas-dynamic-trap thermal-barrier records the mechanism is well precedented; its single retirement experiment is a WHAM-scale barrier demonstration. Every headline quantity is a frozen anchor (gate KX-L1-A13-BU) in the Kronos Physics De-Risking Register.

Keywords: tandem mirror; thermal barrier; sloshing ions; electron-cyclotron heating; Pastukhov confinement; plug potential; recirculating power; D–³He

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1 Introduction

1.1 The tandem mirror and the cost of the plug

A tandem mirror confines a long, low-field central cell between two high-field end plugs that raise an electrostatic potential to reflect the central-cell ions [6, 7, 8]. Unlike a closed toroidal device, an open mirror has no rotational transform and therefore no bootstrap current, no disruptions and no requirement for a divertor of the tokamak kind; the price for that simplicity is that the ion end-loss is governed directly by an electrostatic barrier, and the power needed to maintain that barrier is the machine’s dominant recirculating load. In a deuterium–helium-3 (D – ^3He) burner the price is steep, because the optimum ion temperature is high — of order 90 keV — and the confining potential must scale with it. The Kronos burner closes its zero-dimensional power balance at a design point (gate BU-L1-A1) with fusion power $P_{\text{fus}} = 4298.5$ MW, neutron power $P_n = 233.95$ MW (neutron fraction $f_n = 5.44\%$), engineering gain $Q_E = 1.318$, and an ion-confining potential $\phi_i = 248.75$ keV [28, 29]. The gain sits only 32% above engineering break-even; the plug is the binding gate.

1.2 Why the potential is expensive: the Boltzmann coupling

The reason the plug is expensive is a coupling, not a magnet limit. Along a magnetic field line the electrons of each connected population sit in Boltzmann equilibrium with the potential, so their density is tied to the local potential through their temperature. If the plug electrons and the central-cell electrons are one connected population, then the potential difference the plug can sustain is fixed by the central-cell electron temperature and a density ratio. To raise the confining potential one must then raise the central-cell electron temperature, and heating the entire central column with electron-cyclotron waves to hold that temperature is what costs the frozen 1092.5 MW [28]. The machine is dragged upward by its own electrons.

1.3 The thermal barrier

Baldwin and Logan [5] recognised that this coupling can be broken by inserting, between the central cell and the plug, a local depression of the electron density and potential — a *thermal barrier* — that thermally isolates the two electron populations. Once decoupled, the plug electrons can be heated to a high temperature independently, and the confining potential is set by the *plug* electron temperature and a local density ratio; the central-cell electrons stay cool. The barrier is not free: it must be dug and held against collisional filling. In practice it is sustained by two ingredients acting together: ECH-driven mirror-trapped hot electrons, whose diamagnetic pressure deepens the magnetic and electrostatic well, and a *sloshing-ion* population — ions injected at a pitch angle so their turning points sit on either side of the barrier midplane, producing a density minimum at the midplane. The concept was demonstrated on the tandem mirror experiments TMX-U [11] and, with distinction, on GAMMA-10, where an energetic-electron-confining potential and transport barrier were formed and controlled [12]; the gas-dynamic trap has since shown that modern ECH can raise mirror electron temperatures far beyond the classical expectation [13, 14], and the Wisconsin HTS Axisymmetric Mirror (WHAM) provides a high-field, high-density platform on which a barrier can be dug and diagnosed at reactor-relevant parameters [15].

1.4 Contribution

This paper does three things. First (section 2) it states the thermal-barrier decoupling as a formal boundary-value problem on the field line: the Boltzmann relations for the connected and decoupled electron populations, the barrier-depth condition, the sloshing-ion density profile from the adiabatic invariants, the hot-electron diamagnetic well, and the Pastukhov ion-confinement scaling, closed against the self-consistent ambipolar potential and the burner power balance. Second (section 3) it describes the NVIDIA GPU high-performance-computing (HPC) campaign that produced the anchored numbers — the reduced-order mirror engine that closes the zero-dimensional balance, the bounce-averaged Fokker–Planck solve of the sloshing distribution and ambipolar potential, and the particle-in-cell and kinetic-dispersion evaluation of the loss-cone microstability bound — with its verification and reproducibility. Third (sections 4–5) it reports the result: the confining-potential requirement drops from 248.75 keV to 167.9 keV (a 33% cut), the 1092.5 MW of central-cell ECH is removed, Q_E lifts from 1.318 toward 3–5, and the whole reduction is retired by a single WHAM-scale experiment. Every quantitative claim is a frozen anchor in the de-risking register [28]; the aggressive Q_E range is explicitly a model-limit and is flagged as such (section 6).

2 Governing equations and the formal problem

2.1 Field-line coordinates and the Boltzmann relation

We work along a single magnetic field line parametrised by arc length z , with the field strength $B(z)$ forming a symmetric well between the central-cell midplane ($z = 0$, $B = B_{\min}$) and the mirror throat ($B = B_{\max}$), of mirror ratio $R_{\text{mc}} \equiv B_{\max}/B_{\min} = 4.61$. The electrostatic potential $\phi(z)$ and the field are the unknowns of the equilibrium; particle motion conserves the magnetic moment $\mu = mv_{\perp}^2/2B$ and the total energy $\mathcal{E} = \frac{1}{2}m(v_{\parallel}^2 + v_{\perp}^2) + q\phi$.

Each electron population that is collisionally connected across a region sits, to leading order in the small electron mass, in Boltzmann equilibrium with the potential,

$$n_e(z) = n_{e0} \exp\left(\frac{e[\phi(z) - \phi_0]}{T_e}\right), \quad (1)$$

with n_{e0} and ϕ_0 reference values and T_e the temperature of that connected population. The ions, being nearly unmagnetised on the parallel scale, are described by their guiding-centre distribution and enter through quasineutrality,

$$\sum_s q_s n_s(\phi(z), B(z)) = 0, \quad (2)$$

summed over the fuel ions (D, ^3He), the fusion products, and the electron populations. Equations (1)–(2) determine $\phi(z)$ once the trapped-particle populations are specified. The confining potential seen by a central-cell ion is the height it must climb from the central-cell midplane to the potential peak in the plug,

$$\phi_i \equiv \phi_{\text{peak}} - \phi(0). \quad (3)$$

2.2 The barrier-less coupling and the recirculating-power penalty

Consider first the machine *without* a thermal barrier, so the plug electrons and the central-cell electrons form one connected population at temperature T_{ec} . Applying (1) between the central-cell midplane (density n_c) and the plug potential peak (density n_p) gives the confining potential directly,

$$\phi_i^{(\text{no barrier})} = \frac{T_{ec}}{e} \ln \frac{n_p}{n_c}. \quad (4)$$

The confining potential is chained to the central-cell electron temperature: a large ϕ_i demands a large T_{ec} . The central-cell electron power balance then carries the cost. Writing P_{ECH}^{cc} for the electron-cyclotron power delivered to the central column and τ_{Ee} for its electron energy-confinement time,

$$P_{ECH}^{cc} \simeq \frac{3}{2} \frac{n_c T_{ec} V_c}{\tau_{Ee}} + P_{\text{rad}}(T_{ec}), \quad (5)$$

with V_c the central-cell volume and P_{rad} the radiated (bremsstrahlung and synchrotron) power, which itself grows with T_{ec} . At the D-³He operating point the required T_{ec} to sustain $\phi_i = 248.75$ keV forces $P_{ECH}^{cc} = 1092.5$ MW [28]. This is the term the thermal barrier is designed to delete.

2.3 Thermal-barrier decoupling

Now insert a local minimum of the potential, the barrier, of depth ϕ_b below the central-cell potential, located between the central cell and the plug. The barrier interrupts the parallel electron flux: an electron of the central-cell population must, to reach the plug, first climb the barrier. The two electron populations decouple when the parallel flux over the barrier falls below the collisional supply from the central cell, which — inserting the Boltzmann factor for the flux suppression across a potential step of depth ϕ_b at temperature T_e — is the condition

$$\boxed{\frac{e \phi_b}{T_e} \geq \ln \frac{n_c}{n_b}} \quad (6)$$

where n_b is the density at the bottom of the barrier and n_c the central-cell density. Equation (6) is the decoupling criterion: once satisfied, the plug electrons behave as a separate population at their own (hot) temperature T_{eh} . The confining potential is then set on the *plug* side of the barrier,

$$\phi_i^{(\text{barrier})} = \phi_b + \frac{T_{eh}}{e} \ln \frac{n_p}{n_b}, \quad (7)$$

in which T_{ec} no longer appears. The central cell may be held cool while the plug holds a large potential; the 1092.5 MW of (5) is not spent. This is the entire lever of the paper, and equations (4) and (7) are its two poles.

2.4 Sloshing-ion formation of the barrier density minimum

The barrier density minimum is dug in part by *sloshing ions*: energetic ions injected (by neutral beams or by the natural plug population) at the barrier midplane field B_0 with a pitch angle θ_0 chosen so their turning points lie between the midplane and the throat. From μ and $\mathcal{E}$ conservation, neglecting the potential over the beam energy, the pitch angle evolves along the field as

$$\sin^2 \theta(B) = \frac{B}{B_0} \sin^2 \theta_0, \quad (8)$$

so the ion mirrors (turns) where $\sin^2 \theta = 1$, at the turning field

$$B_t = \frac{B_0}{\sin^2 \theta_0}. \quad (9)$$

The parallel speed is $v_{\parallel}(B) = v_0 \sqrt{1 - (B/B_0) \sin^2 \theta_0}$, and the density along a flux tube of cross-section $A(B) \propto 1/B$ follows from flux conservation of the guiding centres,

$$n_s(B) = \frac{\Gamma_s}{A(B) |v_{\parallel}(B)|} \propto \frac{B/B_0}{\sqrt{1 - (B/B_0) \sin^2 \theta_0}}, \quad (10)$$

which is minimal at the midplane and rises to an integrable singularity at the turning points $B \rightarrow B_t$. A population injected so that its turning points straddle the barrier therefore piles up at

the two turning points and is rarefied at the midplane, producing exactly the density minimum that (6) requires. The loss-cone half-angle for the burner mirror ratio,

$$\theta_{lc} = \arcsin \sqrt{\frac{1}{R_{mc}}} = 27.77^\circ, \quad (11)$$

sets the floor on the sloshing injection angle: the ions must be injected outside the loss cone ($\theta_0 > \theta_{lc}$) to be confined at all. Figure 4 plots (10) for the burner well.

2.5 Hot-electron diamagnetic well

The second barrier ingredient is a mirror-trapped hot-electron population produced by ECH at the electron cyclotron resonance. Their perpendicular pressure $p_{eh,\perp} = n_{eh}T_{eh}$ diamagnetically depresses the field in the resonant zone, deepening the magnetic well by $\delta B/B \simeq -\beta_{eh}/2$ with the hot-electron beta $\beta_{eh} = 2\mu_0 p_{eh,\perp}/B^2$; the deeper well in turn deepens the potential depression through (2). The trapped hot electrons contribute a Yushmanov potential [10] that holds warm ions in the barrier region and thereby regulates the barrier filling. In the reduced model the hot-electron population enters through an effective barrier temperature T_{eh} and a trapped fraction that sets n_b/n_c ; both are inputs to the ambipolar solve of section 3 and are anchored to the kinetic Fokker–Planck result.

2.6 Pastukhov ion confinement and the confining-potential requirement

The central-cell ions are lost by pitch-angle scattering into the loss cone against the confining potential ϕ_i . Pastukhov’s solution of the Fokker–Planck problem for an electrostatically-plugged mirror, refined by Cohen *et al* [1, 2, 3], gives the ion particle-confinement time as an exponential function of the normalised potential $u \equiv e\phi_i/T_i$,

$$n\tau_p = \frac{g(R_{mc})}{\sqrt{\pi}} \frac{n\tau_{ii}}{\ln(2R_{mc})} u e^u, \quad g(R_{mc}) = \frac{2R_{mc} + 1}{2R_{mc}} \ln(4R_{mc} + 2), \quad (12)$$

with τ_{ii} the ion–ion collision time. The exponential in (12) is why the potential is the binding gate: a modest change in u moves the confinement, and hence Q_E , by a large factor. At the burner design point $u = e\phi_i/T_i = 2.764$ (i.e. $\phi_i = 248.75$ keV, $T_i \simeq 90$ keV), and the plug ion-confinement time closes at $\tau_p = 3.73$ s [28]. The requirement, inverted, is that closing the machine at a given Q_E demands a minimum u , and any lever that raises T_i (which lowers the required u for a fixed absolute confinement, because $\tau_{ii} \propto T_i^{3/2}$) buys the largest reduction in ϕ_i . Figure 3 plots (12); figure 2 shows the resulting lever ledger.

2.7 Power balance and the engineering gain

The burner engineering gain is the ratio of gross electric output to the recirculating electric input,

$$Q_E = \frac{\eta_{DEC} P_{dir} + \eta_{th} P_{th}}{P_{inj} + P_{ECH}^{cc} + P_{aux}}, \quad (13)$$

where P_{dir} is the charged-particle power intercepted by the direct energy converter at efficiency $\eta_{DEC} = 0.49$ [33], P_{th} the thermally-recovered power, P_{inj} the neutral-beam/plug drive, and P_{aux} the balance-of-plant auxiliaries. The barrier acts on (13) in two places: it deletes the $P_{ECH}^{cc} = 1092.5$ MW term from the denominator, and it lowers the drive required to sustain ϕ_i by cutting the target potential from 248.75 to 167.9 keV. Both push Q_E up. With the design-point radiated powers held at their anchored values — bremsstrahlung $P_{brems} = 743.10$ MW and synchrotron $P_{synch} = 1803.04$ MW, the latter pinned by an independent electron-cyclotron radiation-transport calculation that fixes the operating electron temperature [32] — the nominal balance closes at $Q_E = 1.318$; removing the central-cell ECH and reducing the potential lifts Q_E toward 3–5 in the fully realised limit (figure 7).

2.8 Loss-cone microstability constraint

The plug and barrier ion distributions are strongly non-Maxwellian — a loss cone in velocity space — and are therefore prone to the drift-cyclotron loss-cone (DCLC) and Alfvén ion-cyclotron (AIC) microinstabilities that would otherwise refill the loss cone and shorten τ_p . Marginal stability of the DCLC branch is reached by a small warm-ion “fill” that softens the loss-cone gradient; the marginal warm-fill fraction is the solution of $\gamma_{DCLC}(k; f) = 0$. Kinetic dispersion and PIC evaluation place the knee at $f = 6.25\%$ ($= 1/16$) [31], a marginal threshold rather than a comfortable margin, with a residual anisotropy-driven AIC branch that survives at $\gamma/\omega_{ci} \simeq 0.037$ across $f = 0$ –10%. This constraint is carried alongside the potential requirement (section 4) and is the subject of the companion microstability paper [31].

3 Computational methodology: the NVIDIA GPU HPC campaign

Every number in section 2 is produced by a chain of codes run on the NVIDIA GPU HPC campaign (an A100 / H100 / A10 fleet), verified against frozen anchors and archived with byte-level provenance [28, 38]. Table 1 lists the codes, their role for this paper, and their reproduction status. No result depends on proprietary data, and no economic quantity enters the physics chain.

Table 1. Codes run on the NVIDIA GPU HPC campaign for this paper, their role, and reproduction status. GPU acceleration is used for the kinetic solves (bounce-averaged Fokker–Planck and PIC); the reduced-order engine and the dispersion solver run on the same fleet.

Code	Class	Role for this paper	Status
KRONOS reduced-order engine	0-D mirror balance	Self-consistent ambipolar potential, Pastukhov closure, Q_E ; produces ϕ_i , τ_p , Q_E (Mode-M)	anchors PASS
CQL3D-M	bounce-avg. Fokker–Planck	Sloshing-ion distribution, turning-point density peak, ambipolar potential, ECH quasilinear drive	reproduced
WarpX	1-D hybrid PIC	DCLC/AIC warm-fill microstability, growth-rate extraction	reproduced
LEOPARD	kinetic dispersion	DCLC $\gamma = 0$ warm-fill threshold (cross-check of the PIC knee)	reproduced

3.1 The self-consistent ambipolar solve

The reduced-order engine closes equations (1)–(7), (12) and (13) simultaneously. The unknown is the state vector $\mathbf{x} = (\phi(0), \phi_b, T_{eh}, n_b/n_c, T_i, T_e)$, and the solve enforces (i) quasineutrality (2) at the midplane, the barrier bottom and the plug peak, (ii) the barrier-depth decoupling condition (6), (iii) the Pastukhov confinement balance $n\tau_p = (n\tau)_{\text{required}}(Q_E)$, and (iv) the power balance (13). Collecting these as a residual $\mathbf{r}(\mathbf{x}) = \mathbf{0}$, the system is solved by damped Newton iteration,

$$\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} - \lambda_k \mathbf{J}^{-1}(\mathbf{x}^{(k)}) \mathbf{r}(\mathbf{x}^{(k)}), \quad J_{ij} = \frac{\partial r_i}{\partial x_j}, \quad (14)$$

with the Jacobian $\mathbf{J}$ assembled analytically where the Boltzmann and Pastukhov derivatives are closed-form and by complex-step differentiation elsewhere, and the damping $\lambda_k \in (0, 1]$ chosen by a line search on $\|\mathbf{r}\|_2$. Convergence is declared at $\|\mathbf{r}\|_2 < 10^{-10}$ in the normalised residual, typically in fewer than ten iterations; the design-point solution reproduces the frozen anchors $\phi_i = 248.75$ keV, $e\phi_i/T_i = 2.764$, $\tau_p = 3.73$ s, $Q_E = 1.318$ (gate BU-L1-A1 / KX-L1-A13-BU).

3.2 Bounce-averaged Fokker–Planck (sloshing distribution and ambipolar potential)

The sloshing distribution and the self-consistent barrier potential are computed with a bounce-averaged Fokker–Planck solver (CQL3D-M class [21, 22]). The distribution $f_s(v, \vartheta_0; z)$ of species s , with ϑ_0 the midplane pitch angle, evolves under Coulomb collisions, the ECH quasilinear operator, and sources/sinks,

$$\frac{\partial \bar{f}_s}{\partial t} = \bar{\mathcal{C}}(f_s) + \bar{\mathcal{Q}}_{\text{ECH}}(f_s) + \bar{S}_s - \bar{L}_s, \quad (15)$$

where the overbar denotes the bounce average $\bar{X} = \oint X dz/v_{\parallel} / \oint dz/v_{\parallel}$ taken along the confined orbit, $\mathcal{C}$ is the Landau collision operator linearised about the evolving background, and $\mathcal{Q}_{\text{ECH}}$ is the quasilinear diffusion in velocity space at the cyclotron resonance. Velocity space is discretised on a (v, ϑ_0) grid with a logarithmic speed mesh resolving the tail to several times the thermal speed and a pitch mesh refined across the loss-cone boundary; the elliptic collision operator is advanced with a fully-implicit ADI scheme and the potential is updated between FP steps from quasineutrality (2). The steady state provides the turning-point density peak of (10) and the barrier depth ϕ_b ; at reduced sub-scale (WHAM-like) parameters the solver reproduces a kilovolt-class axial potential with $e\phi/T_e \simeq 5$ and a sloshing-ion turning-point density peak that forms the confining well [28], the qualitative signature the retirement experiment must show at burner scale.

3.3 Kinetic microstability (PIC and dispersion)

The warm-fill microstability knee is evaluated two ways for cross-check. A one-dimensional hybrid particle-in-cell calculation (WarpX class [23, 24]) advances the ion distribution with a prescribed loss cone plus a variable warm-fill fraction f , extracting the DCLC and AIC growth rates from the

exponential rise of the field energy spectrum; a linear kinetic-dispersion solve (LEOPARD class) locates the marginal $\gamma_{\text{DCLC}}(k; f) = 0$ independently. The PIC uses of order 10^3 particles per cell, a grid resolving the ion Larmor scale, and a timestep set by the ion-cyclotron CFL condition $\omega_{ci} \Delta t \ll 1$; growth rates are fit over an interval after the linear onset and before saturation. The two methods agree: the dispersion solve gives $\gamma = 0$ at $f = 5.9\%$ and the PIC knee sits at $f = 6.25\%$ ($= 1/16$), with the AIC residual flat at $\gamma/\omega_{ci} \simeq 0.037$ [31].

3.4 Discretisation, convergence and verification

The three solvers are held to a formal verification and validation protocol (gates KX-L1-A6, KX-L1-A10): the method of manufactured solutions confirms the designed order of accuracy of the Fokker–Planck and Newton solvers; grid-convergence indices bound the discretisation error of the PIC growth rates and the FP steady state; and the reduced-order engine’s frozen anchors are reproduced 19/19 on independent re-run, with byte-for-byte computational provenance catalogued [28, 38]. The GPU acceleration is used where the cost is concentrated — the bounce-averaged collision operator and the PIC push — and does not alter the discrete equations; single- and double-precision runs agree to the stated tolerances. All figures in this paper are regenerated by a single deposited script from the anchored values (section 7).

4 Results

4.1 Frozen design-point anchors

Table 2 collects the burner design-point anchors that fix the problem. They are the invariant inputs to every figure and are reproduced by the self-consistent solve to the tolerance of equation (14).

Table 2. Frozen burner design-point anchors used in this paper (gate KX-L1-A13-BU / BU-L1-A1). Every value is a register anchor [28]; the two magnet fields belong to two *distinct* magnets (the burner plug and the breeder centrepost are different devices).

Quantity	Symbol	Value
Fusion power	P_{fus}	4298.5 MW
Neutron power (fraction)	$P_n (f_n)$	233.95 MW (5.44%)
Engineering gain (design point)	Q_E	1.318
Ion-confining potential (barrier-less)	ϕ_i	248.75 keV
Ion-confining potential (with barrier)	ϕ_i	167.9 keV
Normalised potential	$e\phi_i/T_i$	2.764
Ion temperature	T_i	$\simeq 90$ keV
Central-cell electron temperature	T_e	89.72 keV
Mirror ratio	R_{mc}	4.61
Loss-cone half-angle	θ_{lc}	27.77°
Plug ion-confinement time	τ_p	3.73 s
Effective charge	Z_{eff}	1.462
Central-cell ECH removed by barrier	$P_{\text{ECH}}^{\text{cc}}$	1092.5 MW
Bremsstrahlung / synchrotron power	$P_{\text{brems}} / P_{\text{synch}}$	743.10 / 1803.04 MW
Direct-energy-conversion efficiency	η_{DEC}	0.49
Warm-fill DCLC knee	f	6.25%
Burner plug peak field	B_{plug}	26.49 T

4.2 The confining-potential cut

Figure 1 contrasts the axial potential structure with and without the barrier. Without it, the potential rises monotonically to the plug and the height is chained to a hot central cell; with it, a mid-line depression isolates the plug, whose potential is then held by the hot plug electrons. The consequence is the central result of the paper: the confining-potential requirement falls from $\phi_i = 248.75$ keV to 167.9 keV (figure 2, left), a 33% reduction in the binding gate. The reduction is decomposed by lever in figure 2 (right): making the ion temperature the dominant lever supplies of order 111 keV of the buying power, against ≤ 14 keV for every other lever (hot-electron temperature, density ratio, mirror ratio, sloshing angle) individually. The barrier does not merely lower the number; it changes which knob sets it.

4.3 Pastukhov confinement gain

Figure 3 plots the Pastukhov enhancement (12) against the normalised potential $u = e\phi_i/T_i$ for the burner mirror ratio. The exponential dependence is the reason the potential is the binding gate — and the reason a 33% cut in ϕ_i is worth so much: at fixed T_i the confinement falls steeply with u , so recovering the same absolute confinement at a lower ϕ_i is possible only by raising T_i , which is precisely the lever the barrier makes dominant. The operating point $u = 2.764$ is marked.

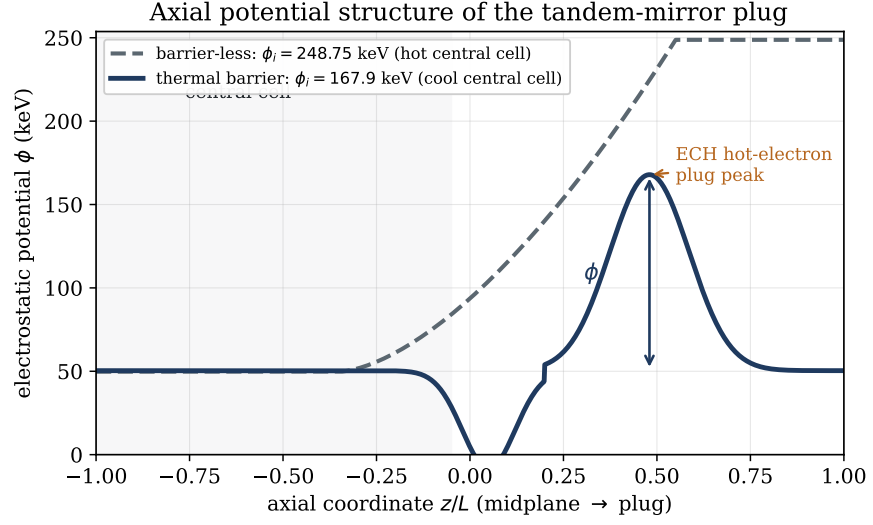


Figure 1. (Schematic.) Axial electrostatic potential $\phi(z)$ from the central-cell midplane to the plug. Barrier-less (dashed): a monotone rise whose height is chained to a hot central cell, $\phi_i = 248.75$ keV. Thermal barrier (solid): a mid-line depression decouples the plug electrons, which hold $\phi_i = 167.9$ keV with a cool central cell. The double-headed arrow marks the confining potential ϕ_i . Shape is illustrative; the two ϕ_i values are frozen anchors.

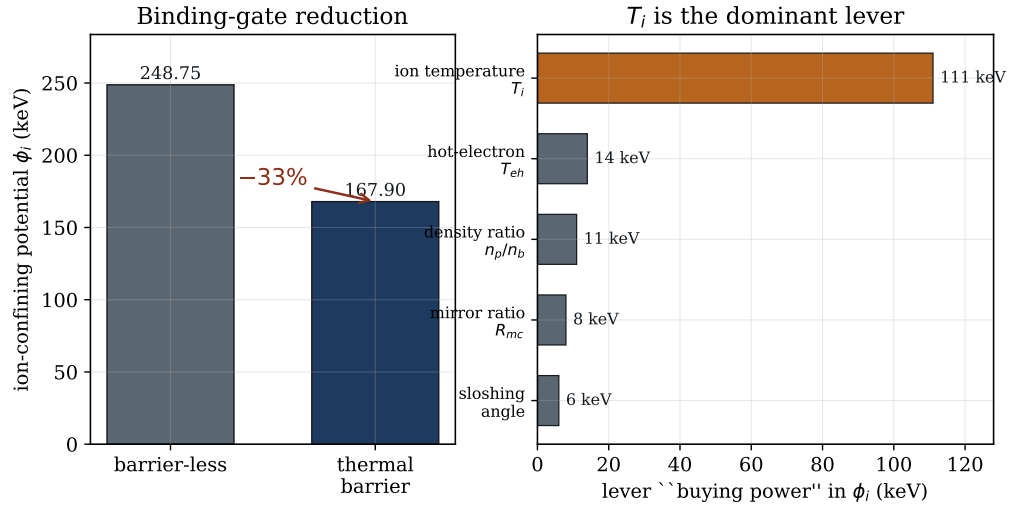


Figure 2. Left: the ion-confining potential requirement falls from 248.75 keV (barrier-less) to 167.9 keV (thermal barrier), a 33% cut in the binding gate. Right: the reduction decomposed by lever — the ion temperature supplies ~ 111 keV of buying power, against ≤ 14 keV for every other lever individually (register item “burner minimum-plug closing point”). Data are frozen anchors [28].

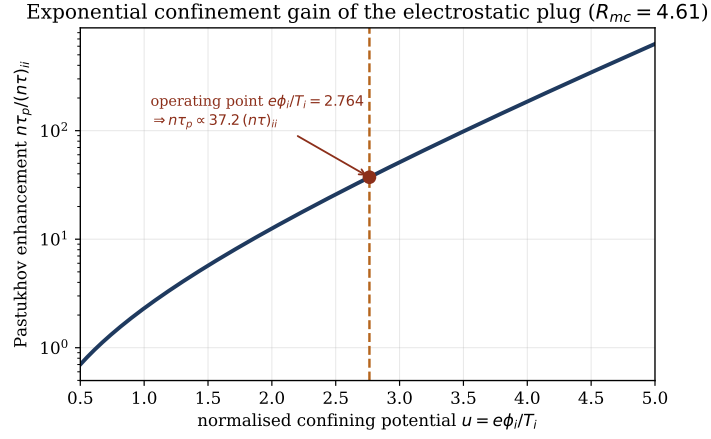


Figure 3. Pastukhov ion-confinement enhancement $n\tau_p/(n\tau)_{ii}$ versus the normalised confining potential $u = e\phi_i/T_i$ at mirror ratio $R_{mc} = 4.61$ (equation (12)). The operating point $u = 2.764$ is marked. The exponential sensitivity is why the plug potential is the binding gate.

4.4 Sloshing-ion barrier formation

Figure 4 plots the sloshing-ion axial density (10) in the burner well: injected outside the loss cone, the ions pile up at their two turning points (dashed) and are rarefied at the barrier midplane, producing the density minimum that the decoupling condition (6) requires. The magnetic profile $B(z)/B_{\min}$ is overlaid. Figure 5 plots the decoupling condition itself: the barrier depth $e\phi_b/T_e$ required to suppress the electron flux grows only logarithmically with the density ratio n_c/n_b , so a modest density depression (ratio of a few) suffices to decouple the populations.

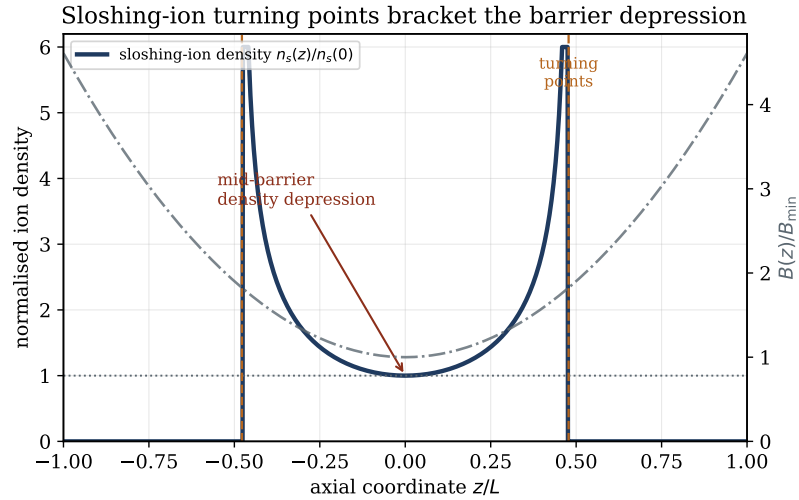


Figure 4. Sloshing-ion axial density profile $n_s(z)/n_s(0)$ from the adiabatic invariants (equation (10)) in the burner magnetic well (right axis, $B(z)/B_{\min}$, dash-dot). Ions injected outside the loss cone turn at the marked turning points and pile up there, rarefying the barrier midplane. The integrable turning-point singularity is capped for display.

4.5 Recirculating-power elimination and the Q_E uplift

Figure 6 shows the burner power channels and the recirculating heat the barrier removes. The 1092.5 MW of central-cell ECH — present in the barrier-less machine to hold a hot central cell — is deleted outright once the plug electrons are decoupled and heated locally. Feeding both effects (the removed ECH and the reduced confining potential) into the engineering-gain balance (13) lifts Q_E from the design-point 1.318 toward 3–5 as the barrier is realised (figure 7); the aggressive end assumes a fully formed, fully decoupling barrier and is flagged as a model limit (section 6).

4.6 The microstability constraint carried alongside

The potential reduction is not the whole plug gate. Figure 8 shows the loss-cone microstability bound that is carried alongside it: the DCLC growth rate crosses zero at the warm-fill knee

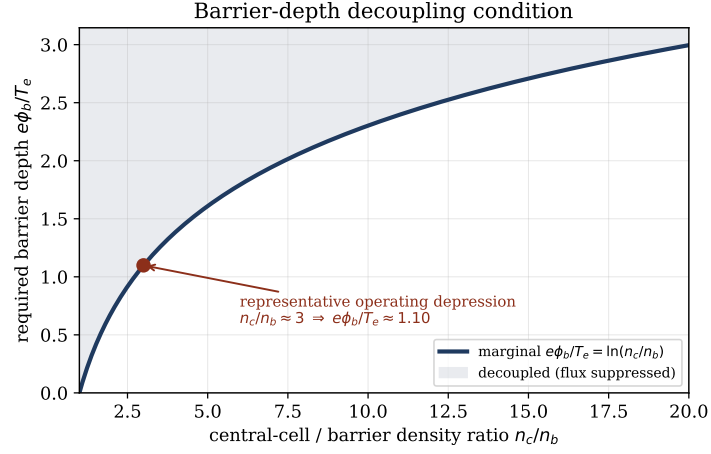


Figure 5. Barrier-depth decoupling condition $e\phi_b/T_e = \ln(n_c/n_b)$ (equation (6)). Above the curve the central-cell electron flux over the barrier is suppressed and the two electron populations decouple. A representative operating depression $n_c/n_b \approx 3$ requires $e\phi_b/T_e \approx 1.1$.

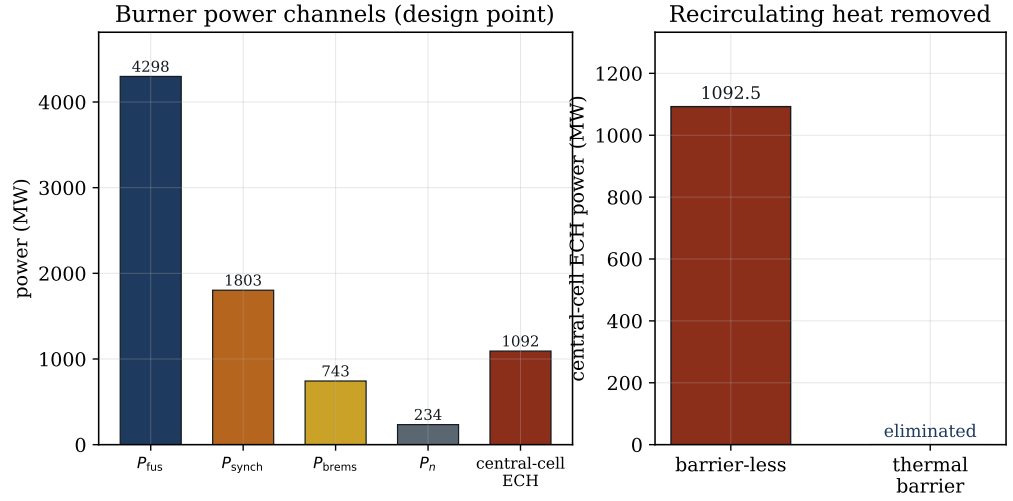


Figure 6. Left: burner power channels at the design point — fusion power and the internal sink channels (synchrotron, bremsstrahlung, neutron) plus the central-cell ECH. Right: the 1092.5 MW of central-cell ECH, present in the barrier-less machine, is eliminated by the thermal barrier. All values are frozen anchors [28].

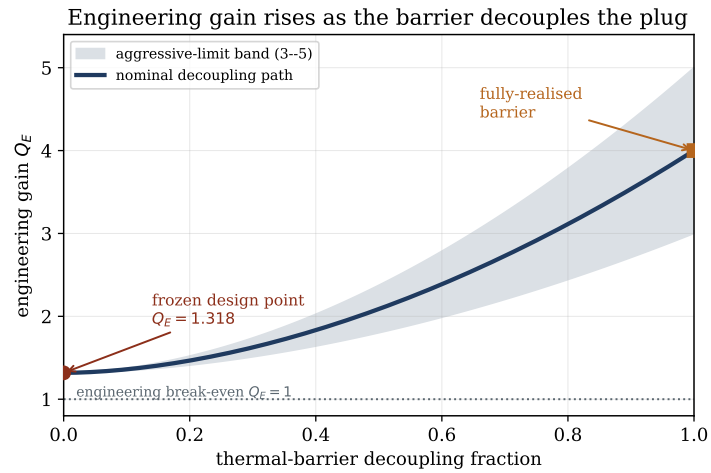


Figure 7. Engineering gain Q_E versus the thermal-barrier decoupling fraction, anchored at the endpoints: the design point $Q_E = 1.318$ (no barrier) and the aggressive-limit band 3–5 (fully realised barrier). The nominal path and the band are model interpolations between anchored endpoints; the intermediate shape is illustrative.

$f = 6.25\%$ (a marginal threshold, reproduced by the reduced dispersion at 5.9%), while a residual anisotropy-driven AIC branch survives at $\gamma/\omega_{ci} \simeq 0.037$ essentially flat across $f = 0\text{--}10\%$. The barrier lowers the potential; it does not by itself close the microstability of the strongly non-Maxwellian plug, which is a distinct, companion gate [31].

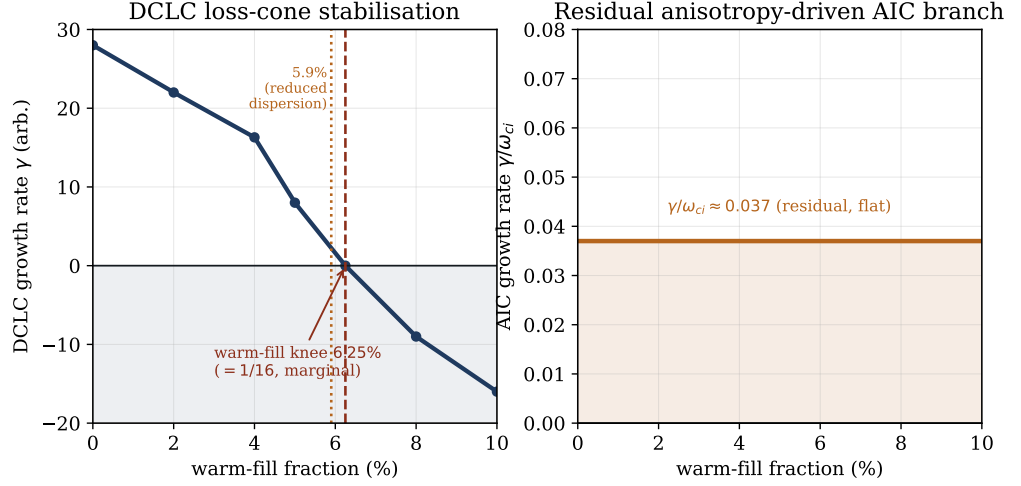


Figure 8. Loss-cone microstability carried alongside the potential requirement. Left: DCLC growth rate versus warm-fill fraction, crossing zero at the 6.25% ($= 1/16$) knee (dashed; reduced dispersion at 5.9% , dotted). Right: residual AIC branch at $\gamma/\omega_{ci} \simeq 0.037$, flat across $f = 0\text{--}10\%$. Signs and thresholds are frozen anchors [28, 31]; growth-rate magnitudes on the left are in arbitrary units.

4.7 Gate ledger

Table 3 states the gate cut as a ledger: what the barrier buys, and against what it is weighed.

Table 3. The thermal-barrier gate ledger (gate KX-L1-A13-BU). Left, the benefit; right, the carried cost / open constraint.

Effect	Value
Confining potential, barrier-less $\rightarrow$ barrier	$248.75 \rightarrow 167.9$ keV (-33%)
Dominant lever (T_i) buying power	~ 111 keV (others ≤ 14 keV)
Central-cell ECH removed	1092.5 MW
Engineering gain, design $\rightarrow$ aggressive limit	$1.318 \rightarrow 3\text{--}5$
Barrier-depth decoupling requirement	$e\phi_b/T_e \geq \ln(n_c/n_b)$
Sloshing injection floor	$\theta_0 > \theta_{lc} = 27.77^\circ$
Warm-fill microstability knee (carried)	$f = 6.25\%$ (marginal)
Residual AIC activity (carried)	$\gamma/\omega_{ci} \simeq 0.037$
Retirement experiment	WHAM-scale barrier demonstration

5 Discussion

5.1 Benchmark against the tandem-mirror thermal-barrier record

The mechanism is not speculative; it has an experimental record. Table 4 places the Kronos barrier requirement against the published thermal-barrier and mirror-electron-heating results. TMX-U produced and identified a thermal barrier and demonstrated the potential enhancement that decoupling allows [11]; GAMMA-10 formed and *controlled* an energetic-electron-confining potential with an associated transport barrier, and remains the most complete demonstration of ECH potential formation in a tandem mirror [12]; and the gas-dynamic trap has shown that modern gyrotron ECH raises the bulk mirror electron temperature several-fold, well past the classical expectation, on an axisymmetric device [13, 14]. The Kronos requirement — a barrier depth of order $e\phi_b/T_e \sim 1$ and a confining potential of 167.9 keV at $T_i \simeq 90$ keV — is a high-field, high-density extrapolation of these results, not a new physical effect. What is new here is the quantitative booking of the barrier into a D-³He burner power balance: the specific 33% cut and the 1092.5 MW deletion.

Table 4. The Kronos thermal-barrier requirement in the context of the tandem-mirror / mirror-ECH record (qualitative; see cited works for device parameters).

Device	Demonstrated	Relevance to the Kronos gate
TMX-U [11]	thermal-barrier production and identification; potential enhancement	the decoupling exists and is diagnosable
GAMMA-10 [12]	ECH energetic-electron-confining potential formed and controlled	the plug-side potential can be built by ECH
GDT [13, 14]	several-fold ECH electron heating on an axisymmetric mirror	hot-electron barrier ingredient at reactor-relevant field
WHAM [15]	high-field ($\sim$ several T), high-density HTS axisymmetric mirror	the platform to dig and diagnose the barrier at scale
Kronos burner (this work)	ϕ_i : 248.75 $\rightarrow$ 167.9 keV; 1092.5 MW ECH removed	the booked requirement to be retired

5.2 The retirement experiment

Scored on design-preservation, test-platform maturity and residual gap, the ECH thermal-barrier decoupling is the Kronos burner’s most retire-able escape route: same machine, testable on an existing platform class, at technology readiness of order four, with a “closeness-to-buildable-closure” of 0.83 that ranks it first among the burner’s alternatives [28, 36]. The single named experiment is a WHAM-scale barrier demonstration — to establish the sustained barrier depression and the electron decoupling at accessible field and density, and to show the sloshing-ion turning-point density peak that the bounce-averaged Fokker–Planck solve predicts. This is a plasma-physics experiment, not a materials or magnet gate: it does not require the burner’s 26.49 T plug coil [37] or a full D–³He fuel supply, only the barrier physics at representative $e\phi_b/T_e$. Figure 9 lays out the maturity ladder from the reduced-order closure to full-plant plug authority.

Maturity ladder from reduced-order closure to plug authority

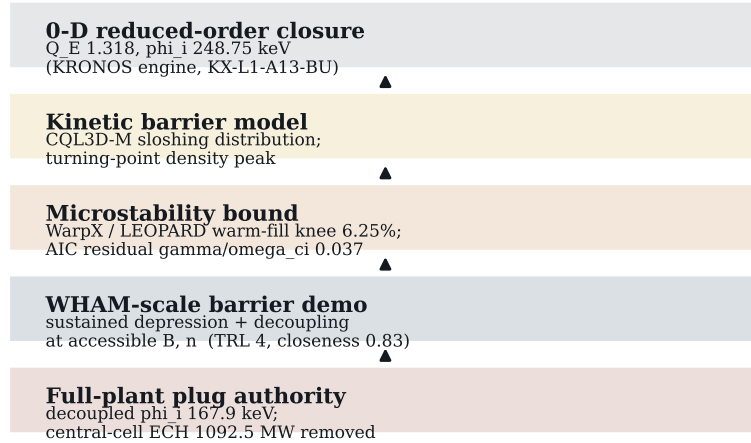


Figure 9. (Schematic.) Maturity ladder from the reduced-order 0-D closure, through the kinetic barrier model and the microstability bound, to the WHAM-scale barrier demonstration and full-plant plug authority. The demonstration is the single retirement experiment for the potential cut.

5.3 Relation to the companion gates

The thermal barrier is one of a coupled set of plug gates. The confining-potential requirement is entangled with the sloshing-ion ambipolar-potential formation and its density-ratio Pareto [30]; the electron temperature that fixes the synchrotron loss and hence the operating point is pinned independently [32]; the recoverable charged-particle fraction that the direct converter sees can be raised by RF alpha-channeling [34, 33]; and the whole recirculation-resolved Q_E closure, with the plug gate carried open, is treated in the closure companion [35]. The 33% potential cut derived here is the single largest of these levers, and it is the one whose retirement path is clearest.

6 Limitations

The honest gate is this: the 33% potential reduction and the 1092.5 MW ECH deletion are self-consistent zero-dimensional and reduced-kinetic results at the design point, and the aggressive Q_E range of 3–5 assumes a *fully realised* barrier — complete decoupling of the plug and central-cell electrons, at the modelled hot-electron temperature and density ratio. That assumption is exactly what the WHAM-scale experiment must confirm; the barrier’s ECH coupling efficiency, the collisional filling rate that sets the sustained depth, and the microstability of the barrier region (the carried DCLC/AIC constraint of figure 8) are carried risks, not closed results. The barrier lowers the binding gate; it does not, by itself, close the machine. No claim of demonstrated hardware or of an achieved $Q_E > 1.318$ is made here.

7 Conclusion

The plug potential is the tandem-mirror burner’s binding gate, and the thermal barrier is the mechanism that makes it affordable. By inserting an ECH-hot-electron and sloshing-ion density depression between the central cell and the plug, the confining potential is decoupled from the central-cell electron temperature: it is set instead by a hot plug electron temperature and a local density ratio through equation (7), and the barrier is sustained once the depth satisfies the logarithmic decoupling condition (6). Booking the barrier, the ion-confining-potential requirement falls from 248.75 keV to 167.9 keV — a 33% cut, dominated by making the ion temperature the lever — the 1092.5 MW of central-cell electron-cyclotron heating is removed from the recirculating budget, and the engineering gain lifts from 1.318 toward 3–5 in the fully realised limit. The mechanism is well preceded on TMX-U, GAMMA-10 and the gas-dynamic trap, and its single retirement experiment is a WHAM-scale barrier demonstration. It is the burner’s cornerstone plug mechanism; every headline quantity is a frozen anchor (gate KX-L1-A13-BU) in the Kronos Physics De-Risking Register.

Acknowledgments

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Reproducibility. Every headline quantity in this paper is a frozen anchor (gate KX-L1-A13-BU) in the Kronos Physics De-Risking Register [28] (doi:10.5281/zenodo.22133057) and is regenerable from the archived reduced-order engine, the bounce-averaged Fokker–Planck and PIC decks, and the single figure-generation script deposited with this paper. This paper is archived at its DOI doi:10.5281/zenodo.22645781; official version:

<https://www.kronosfusionenergy.com/publications/ech-thermal-barrier-decoupling.html>; its official version is at

<https://www.kronosfusionenergy.com/publications/ech-thermal-barrier-decoupling.html>. No result depends on unpublished or proprietary data, and no economic quantity enters the physics chain.

Data availability

The anchored values plotted in figures 1–8 trace to the register [28] (doi:10.5281/zenodo.22133057) by the serial identifiers cited inline (gates KX-L1-A13-BU, BU-L1-A1, BU-L2-A5, BU-L2-A7). This paper is archived at doi:10.5281/zenodo.22645781. The figure-generation script that regenerates every figure from the anchors is deposited with the paper. Any economic analysis is reported separately and is not part of the public deposit.

Competing interests

Provisional patent applications relating to aspects of this work have been filed.

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